EE3408 **ANALOGUE COMMUNICATIONS**

August 2025

Tutorial 1

1.

(a) Explain why a speech signal cannot be transmitted directly through free space.

(b) Briefly explain why amplification of the received signal to make up for attenuation is of no avail.

(c) What is the theoretical bandwidth of a 1 kHz square wave?

2.

(a) For a signal g(t) shown in Figure 1 below, sketch g(-t).

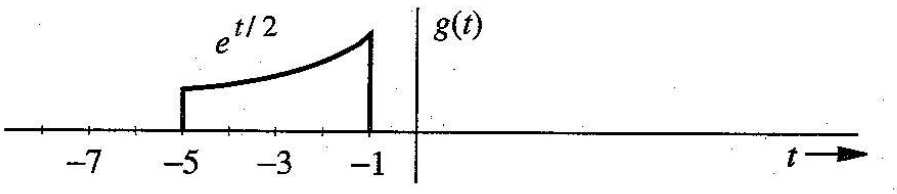


Figure 2.1

(b)

(i) Determine the energy of the signal shown in Figure 2 below.

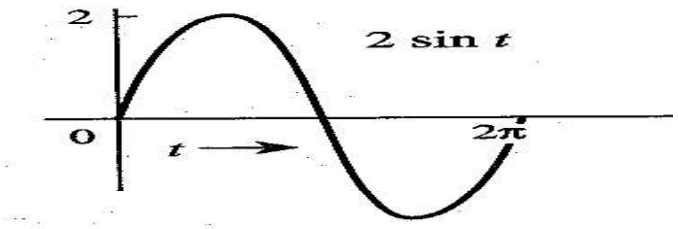


Figure 2.2

(ii) Comment on the effect on the energy of sign change and time shifting of the signal in Figure 2.2.

(iii) What is the effect on the signal energy if the signal in Figure 2.2 is multiplied by K?

(c) Determine the noise current for a diode with a bias current of 15 mA, observed over a 25 kHz bandwidth.

3.

(a) What essential property of a time function determines whether a signal will possess a line spectrum or a continuous spectrum?

(b) Determine the Fourier series of the sawtooth waveform shown in Figure 3.1 below and sketch its spectrum.

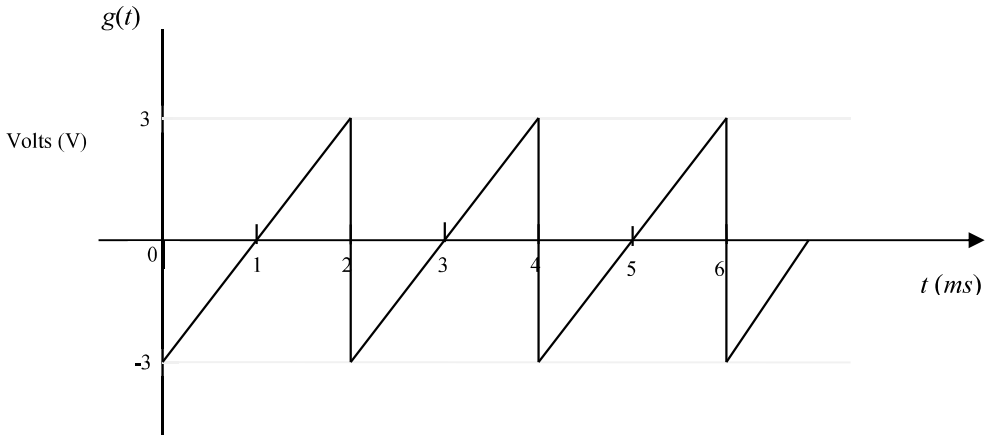
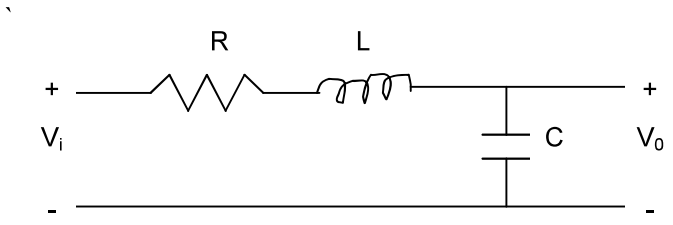


Figure 3.1

4.

Consider the circuit diagram shown below



(a) Determine the transfer function H(s) of this filter.

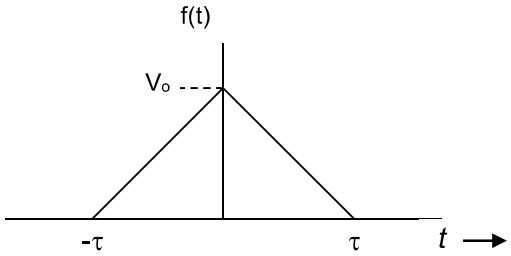
(b) Determine w_o and the Q for this filter.

(c) What type of filter is this?

(d) What is the order of this filter?

5.

(a) Determine the frequency spectrum (Fourier Transform) of a single triangular pulse specified by the conditions:



Where *t* represents time and τ is a constant.

(b) Sketch the spectrum obtained